

What is data ?

collection of information.

What is database ?

collection of data.

What is DBMS ?

DBMS where we perform CRUD operation on our data.

Types of DBMS –

1. **RDBMS** – data store in table
2. **NRDBMS** – data not store in table

What is SQL?

SQL stands for Structured Query Lang. It is a programming lang. used to interact with relational database. It is used to perform CRUD operations.

What are the different types of SQL commands? (DDL, DML, DCL, TCL)

DDL : Data Definition Language – create, truncate, alter, rename, drop

DQL : Data query language – select

DML : Data manipulation language – insert, update, delete

DCL : data control language – grant, revoke,

TCL : Transaction control language – start transaction, commit, rollback

What is the difference between DELETE, TRUNCATE, and DROP?

Delete : Deletes specific rows based on condition

Truncate : Deletes all rows but retains table structure

Drop : Deletes the entire table or database

What are primary keys and foreign keys?

Primary Key : Uniquely identifies each row in a table.

Foreign key : Creates a link between two tables by referencing a primary key in another table.

What is a unique key?

A UNIQUE key ensures that the values in a column are not duplicated across rows in a table.

What is the difference between CHAR and VARCHAR?

Char : Stores fixed-length character strings.

Varchar : Stores variable-length character strings.

What is a database schema?

A database schema is the design that describes how data is stored, organized, and connected in a database.

What is a default constraint in SQL?

A default constraint automatically assigns a predefined value to a column when no value is specified during data insertion.

What is the difference between SQL and NoSQL databases?

SQL databases use structured tables with fixed schemas and strong consistency, while NoSQL databases provide flexible schemas and high scalability for large and unstructured data.

What is the difference between WHERE and HAVING clauses?

Where clause : Filters rows before grouping or aggregation happens.

Having Clause : Filters groups after aggregation is performed.

What is a join in SQL? Explain types of joins.

Joins – Join is used to combine rows from 2 or more tables based on a related columns between them.

Types of join : -

Inner join – returns records that have matching values in both tables.

Left join – returns all records from left table & matched records from right table.

Right join - returns all records from right table & matched records from left table.

Self join – find information in same table.

Full join – returns all records when there is a match in either left or right table.

Cross join – it creates a Cartesian product of two tables, meaning it returns every possible combination of rows from both tables.

What is a subquery?

A subquery or inner query or nested query is a query within another SQL query.

What is a VIEW?

A view is a virtual table that is based on the result of a SQL query.

What are triggers?

A trigger is a database object that automatically executes in response to INSERT, UPDATE, or DELETE operations on a table

What is union and union all ?

Union :- it combines result & remove duplicates rows.

Union all :- it combines result without removing duplicates rows.

Explain ACID properties.

ACID stands for: Atomicity , Consistency, Isolation , Durability

ACID properties are a set of rules that ensure reliable and consistent database transactions.

SQL clauses :

In – matches values in a list.

Between – range of value

Like – patterns matching

Is null – checks null values

Distinct – removes duplicates

Operators :-

1. Arithmetic – addition, subtraction, multiplication, division, modulus
2. Comparison – equal, not equal, greater than, greater than equal, less than, less than equal,
3. Logical – all, between, in, and, like, or, not, any
4. Bitwise – bitwise and , bitwise or

Normalization : - Normalization in SQL is the process of organizing relational data into well-structured tables to reduce redundancy and ensure data integrity.

Aggregate function : - **count (), max(), min(), sum(), avg()**

CASCADE :- CASCADE ensures that when a record in a parent table is updated or deleted, the related records in the child table are automatically updated or deleted.

ON DELETE CASCADE

ON UPDATE CASCADE

What is the difference between IN and EXISTS?

IN checks whether a value matches any value in a list or the result set of a subquery.

EXISTS checks whether a subquery returns at least one row.

What are indexes in SQL?

An index in SQL is a database object that improves the speed of data retrieval operations on a table, similar to how an index in a book helps you find information faster without scanning every page.

Difference between clustered and non-clustered indexes.

A clustered index defines the physical order of data rows in a table.

A non-clustered index is a separate structure that stores index keys and pointers to the actual data rows.